

Abed Zahed

RF & Wireless Systems Engineer | Telecom & Wireless Sensing | Engineering Lead

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Nationality: Swedish | Location: Sharjah, UAE



EXPERIENCE

Academic Research Project: Satellite RF Link Budget & Modulation Simulation

American University of Sharjah (Research Collaboration)

06/2026 - Present 📍 Sharjah, UAE

- Designed a physics-based link budget model for a GEO S-band satellite link in an ongoing academic research collaboration with the American University of Sharjah; modeled multipath, scintillation, and Faraday rotation referencing ITU-R M.1343 and ECSS-E-ST-50-05C, targeting peer-reviewed publication.
- Implemented CCSDS LDPC adaptive coding and modulation in MATLAB, spanning 3 active QPSK profiles ($r=1/2$ to $r=3/4$) and extending evaluation to 8PSK $r=7/8$ and DVB-S2 16APSK per ETSI EN 302 307-1 for robust link performance across varying channel conditions.
- Ran Monte Carlo simulations generating BER waterfall curves via MATLAB Communications Toolbox; achieved 9.5 dB coding gain vs uncoded QPSK, reducing transmit power by 9x.

Consultant Engineer

Fun Robotics

11/2024 - 08/2025 📍 Dubai, UAE

- Managed end-to-end preparation and implementation of technical training courses, coordinating between Fun Robotics and TDRA (UAE Telecom Regulatory Authority) on logistics, scheduling, and program delivery.
- Designed and delivered technical workshops with TDRA, managing curriculum development and coordination with Etisalat Academy.
- Led a team to develop and deliver technical training on wireless communications and RF systems.

Tech Consultant

Mpya Sci & Tech

10/2020 - 08/2024 📍 Gothenburg, Sweden

- Worked as engineering consultant across Ericsson and Veoneer client sites via Mpya Sci & Tech's professional-services model.
- Held After-Work Committee Coordinator and Consultant Buddy roles at Mpya, driving team culture initiatives and onboarding new consultants.

SUMMARY

RF and wireless systems engineer with 5+ years of experience, including Ericsson (led MRC/MRCx implementation, 2.7 dB signal gain; MATLAB, Python, and FPGA test bench for SISO/XPIC/MIMO) and Veoneer (65% to 98% production turnaround via root cause analysis). M.Sc. Wireless, Photonics and Space Engineering, Chalmers; B.Sc. Electrical Engineering, AUS; 4 IEEE publications in RF sensing.

LANGUAGES

Arabic	● ● ● ● ●	Native
English	● ● ● ● ●	Full Professional
Swedish	● ● ● ● ●	Elementary (improving)
Spanish	● ● ● ● ●	Elementary

LICENSES

- ✓ EU (Sweden) Driving License: Category B
- ✓ UAE Driving License

Microwave DSP Config Developer (Consultant)

Ericsson

03/2022 - 12/2023 📌 Gothenburg, Sweden

- Implemented MRC and novel MRCx combining on XPIC channel modes for Ericsson MiniLink; achieved 2.7 dB signal power gain.
- Developed MATLAB and Python test bench for MiniLink validating SISO, XPIC, and MIMO channel configurations end-to-end.
- Extended modulation profile work beyond MRC/MRCx combining to cover higher-order QAM schemes up to 32-QAM.

Process Engineer (Consultant)

Veoneer / Magna International

11/2020 - 01/2022 📌 Vårgårda, Sweden

- Improved machine performance from 65% to 98% by leading a team of 3 optics engineers in troubleshooting, root cause analysis, and documentation, part of the camera production line's industrialization and ramp-up to full-scale manufacturing.
- Directed incident and problem resolution for camera production; coordinated cross-functional investigations with production, engineering, and maintenance teams.
- Built Power BI dashboards and Python pipelines to track production efficiency and process capability KPIs; adopted org-wide for yield monitoring.

Space Systems Research (M.Sc. Program)

Onsala Space Observatory / Chalmers University

2018 - 2019 📌 Onsala, Sweden

- Completed research training at Onsala Space Observatory covering satellite communications, link budgets, and space systems engineering as part of M.Sc. in Wireless, Photonics and Space Engineering at Chalmers.
- Studied orbital mechanics and satellite platform design, radar systems with Doppler processing and FMCW waveforms, and satellite positioning/GNSS with pseudorange-based differential positioning.
- Operated GNSS tracking, satellite positioning, and antenna dish control systems in an active deep space observatory, generating the raw datasets used for deep-space data processing and analysis.

SKILLS

CORE COMPETENCIES

Technical Training & Knowledge Transfer

Regulatory Compliance (ETSI / ANSI / IEC)

Cross-functional Team Leadership

TDRA / UAE Telecom Regulatory Coordination

TELECOM & ENGINEERING

Machine Learning (ANN, SVM, RF sensing classification)

SQL (data queries for analysis)

Digital Modulation and Coding (QPSK, 8PSK, DVB-S2, CCSDS LDPC)

Wireless & Microwave Systems (LTE/5G, Radio Links)

RF Testing, Validation & Performance Analysis

RF Lab Equipment and Instrumentation (VNA, Spectrum Analyzer, Signal Generators, Network Analyzers)

Circuit and PCB Design (RF antenna PCBs, fractal antennas, diagnostic test setups; Altium)

RF Front-End Systems (LNAs, PAs, mixers, filters, T/R modules; systems-level design)

Digital Signal Processing (DSP)

MIMO / XPIC / SISO channel modeling

TOOLS & PLATFORMS

Power BI

Python (test automation, data analysis)

MATLAB

Electronics Testing and Diagnostics (RF systems, channel emulation, performance validation)

FPGA-based channel emulation

Simulink

HFSS

ADS, CST Microwave Studio (EM simulation)

Master's Thesis: RF Researcher

RISE Research Institutes of Sweden

02/2019 - 06/2019 📌 Borås / Gothenburg, Sweden

- Increased detection sensitivity by 3x via patch antenna research for transformer partial discharge monitoring.
- Designed and tested a prototype patch antenna using VNA, spectrum analyzer, and signal generators; calibrated test setup demonstrated performance comparable to standardized measurement setups.
- Measured and characterized radiation patterns for the patch antenna prototype; validated gain, return loss, and 50Ω impedance matching across the UHF frequency range.

Robotics Engineer

Fun Robotics

06/2016 - 08/2017 📌 Dubai, UAE

- Taught wireless communication fundamentals, protocol configuration, embedded systems, and RF signal concepts to students age 7-20.
- Designed comprehensive teaching curricula and supervised student project work across robotics and software platforms for school- and university-level students; adopted as educational resources for the academic industry.
- Designed and manufactured robotic solutions as part of the product development team.

Visiting Researcher

American University of Sharjah

01/2015 - 10/2015 📌 Sharjah, UAE

- Co-authored 4 IEEE journal and conference publications in RF sensing and diagnostic antenna design.
- Designed PCB-based RF antennas with 50Ω coaxial feeds; applied transmission line theory for impedance matching and optimized RF circuit feed networks.
- Simulated 4th-order Hilbert fractal antenna designs in HFSS; characterized 2D and 3D radiation patterns to verify semi-spherical coverage and directivity at VHF/UHF (0.3-6 GHz).

EDUCATION

M.Sc. in Wireless, Photonics, and Space Engineering

Chalmers University of Technology

2017-2019 📌 Gothenburg, Sweden

B.Sc. in Electrical Engineering (Minor: Computer Engineering)

American University of Sharjah

2011-2015 📌 Sharjah, UAE

KEY ACHIEVEMENTS

◆ MRC/MRCx Implementation

Led implementation on Ericsson MiniLink products achieving 2.7 dB gain in received signal power, effectively doubling received power.

◆ UAE Telecom Training Delivery

Designed and delivered technical training with UAE TDRA in partnership with Etisalat Academy, supporting regulatory capability building.

◆ Production Performance Turnaround

Improved camera production line performance from 65% to 98% at Veoneer/Magna by leading a team of 3 optics engineers through root cause analysis and corrective action.

◆ RF Sensing Research

Increased detection sensitivity 3x via patch antenna research at RISE Sweden; co-authored 4 IEEE publications in RF sensing and diagnostic antenna design.

◆ Operational Analytics

Developed Power BI dashboards adopted org-wide at Veoneer for production yield monitoring and Python-automated data analysis pipelines.